Actual F13 twist transfer and finite useful S3 comparisons

Quadratic-twist study, PR2381; cm8-transfer-20261025;25 October2026

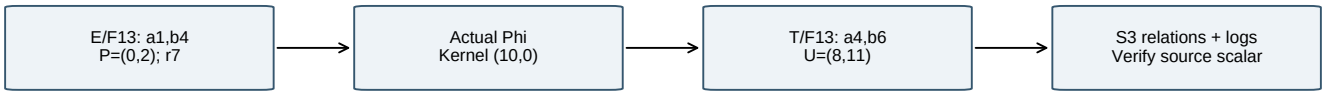
Executed. C++ and independent Rust verify all14 rational source points, 196 addition pairs, the exact two-point kernel, seven scalar images and72 separate main-source recoveries across12 base arms. No point map is constructed by a scalar lookup. All source scalars verify; labels remain outside recovery.

An actual reduction on the order-seven subgroup

Registered E/F13 has $y^2 = x^3 + x + 4$, intermediate H has $y^2 = x^3 + 4x + 7$, and the d2 monic twist T has $y^2 = x^3 + 4x + 6$. The translated degree-two coordinate construction followed by unit5 scaling is

$$X = 12(x + 2/(x-10)), \qquad Y = 8y(1 - 2/(x-10)^2).$$

Infinity and kernel point(10,0) map to infinity. All other source denominators are nonzero. Public lexicographic generator P=(0,2) maps to U=(8,11), both order7. Lean constructs actual nonsingular elliptic points and kernel-checks all subgroup scalar/addition identities. For Q in the selected source subgroup, $\Phi(Q) = nU$ iff $Q = nP$, for every $n \in \mathrm{Fin}\,7$.



No unit c satisfies $c^4 = 4, c^6 = 8$. Lean also checks the necessary coefficient equations for every admissible short-Weierstrass change with parameters u,r,s,t. The general library classification/normalization bridge and generic geometric isogeny theorem are not formalized by these finite checks. Exact ICV1 model identities are retained; selected EC1 generator representations are not added.

Useful rows and named solver degree

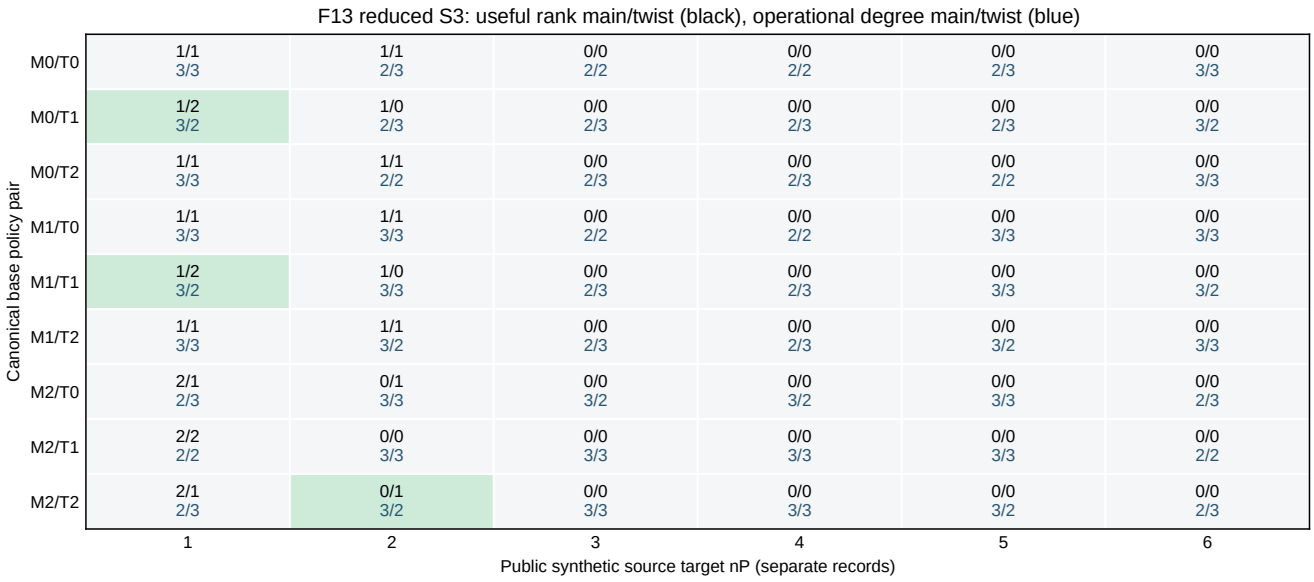
All three size-two bases per curve, all nine canonical pairings and all six nonzero source targets remain:54 paired target comparisons. Transported and pulled-back controls give12 arms,144 original/reduced complete certificates. Every promoted row retains its witness in the three supplied input polynomials.

First source Q=P	Main base(0,7)	Twist base(6,11)
Incremental useful rank	1	2
Ordered constrained roots	1	3
Original operational degree	4	4
Original expanded witness peak	6	4
Reduced operational degree	3	2
Reduced expanded witness peak	3	2

Main first row is(0,2); twist rows are$(0,-2)$,(1,1), independent modulo7. Lean checks the exact actual-point signed-lift sets, determinant and twist base-log solution(5,3). Native setup solves actual relation equations until rank2 and verifies the logs. Recovery receives Q alone, runs the actual reduced solver and retains all ordered sign tests, including those after first success. The public base pullback currently enumerates seven public subgroup points. Native counts retain20setup relation queries,72online queries and36online forward-map calls, without assigning an equal-cost unit or measured time.

Exactly3of54 reduced comparisons have both larger useful rank and lower named operational degree; original S3 has0. Both complete streams reach rank2. The same leading-form cutoff2 holds on both sides. These finite fixed-policy results are not stationary yield, intrinsic regularity, F4/F5 time or a speedup.

All paired targets and the transported-base control



cm8-transfer-20261025: all54 target comparisons; green = native joint higher useful rank/lower degree. No timing or regularity claim.

The graph retains every policy/target. Green identifies exactly the three joint reduced gains. Each cell shows useful-rank main/twist above operational-degree main/twist. The strongest source policy is not replaced by a batch average.

Every transported/pulled-back control preserves the ordered accepted lifts, coefficient rows and useful-rank steps. Of72 matched degree checks,68 equal and 4 differ. At n3,4 main0 to matched0 changes2 to3; twist2 to pullback2 changes3 to2. All four favor the source presentation. Nonlinear encoding changes this solver degree even when matched relation yield is conserved; the counterevidence is retained. No uniform twist advantage follows.

Checked proofs and fail-closed replay

The unmodified scoped gate passes208 public named theorems/269 audited declarations, including17 new public theorems/20 declarations; zero failures or extra trusted axioms. Only propext, Classical.choice and Quot.sound occur. The new universal cleared Montgomery polynomial has source-bound certificate IDC1h45f7f67f60c827c3,32+16 native points and false identity/source controls. It certifies a polynomial only. All16 older identities/sources are unchanged.

Rust independently reproduces field/group arithmetic, roots, incremental Macaulay rows/promotions, evaluation ranks and full certificates, with a separate semantic witness audit. It reproduces24 frozen prefix certificates by actual x; no old scalar labels construct new solver answers. Exact fresh replay and22 backend tests pass. Eleven rehashed controls reject101 without a receipt: false map/subgroup/source, missing arm/target, degree/witness/row/log/query/scalar. There are no failed online shifts, so the missing-failed-query control is explicitly null; missing-query removes a successful query.

Fresh release library4479 tests pass,119 existing ignores; four workflow metadata tests and naming/card/size/artifact checks are recorded with raw source, failures and exact commands. Failed Rust borrow/Lean rewrite attempts and an initial certificate extra-count mismatch remain preserved. No native cap hits. Native budgets600s, two existing build workers, no paid compute or upload. Conductor exits127; no task lease or replacement coordinator is claimed.

Next checkable bridge

Proposed explicit dual, not tested here: with v=12x-6 on T,

$$D_X = 10(v+5+8/v)+10, \qquad D_Y = 12y(1-8/v^2).$$

Handle infinity/kernel(7,0), check both compositions equal doubling, then use $[4]D$ as the order-seven inverse before replacing enumerative base pullback. Expose source-annihilator hypotheses in a generic group theorem. Preserve all prior bases/results and price maps separately; no automatic cost gain is implied. General normalization, geometric maps, broader S3/S4 lifting, stationary useful yield, generic pairing, trace-zero and calibrated complete work remain open. No general hardness, speed, GAE/S/rho or external-fastest claim is admitted.